Clinical Scores and DOAC Master Review

All high-yield clinical scores from every topic (cardio, VTE, neuro, ID, GI/liver, renal, pulmonary, psych, endocrine, rheum, ICU) + complete DOAC guide

PART A — ANTICOAGULATION AND BLEEDING SCORES

1. CHA2DS2-VASc (stroke risk in AF)

Letter	Risk factor	Points
C	Congestive heart failure / LV dysfunction	1
H	Hypertension	1
A2	**Age ≥ 75**	**2**
D	Diabetes mellitus	1
S2	**Stroke / TIA / thromboembolism**	**2**
V	Vascular disease (prior MI, PAD, aortic plaque)	1
A	Age 65–74	1
Sc	Sex category (female)	1
	Maximum	**9**

- **ACC/AHA 2023: OAC recommended** if **≥ 2 (men) or ≥ 3 (women)** • **reasonable** if **1 (men) or 2 (women)** • **0 (men) / 1 (women) → no OAC**.
- **ESC 2024 CHA2DS2-VA** (sex removed): OAC if ≥ 2, consider if 1.
- Female sex alone (score 1) is **not** an indication — it is a risk modifier.
- Applies to paroxysmal, persistent, permanent AF and atrial flutter • **not used** for mechanical valves or moderate–severe mitral stenosis (always anticoagulate with warfarin).

2. HAS-BLED (bleeding risk on anticoagulation)

Letter	Risk factor (definition)	Points
H	**Hypertension** — uncontrolled, **SBP > 160 mmHg**	1
A	**Abnormal renal function** (dialysis, transplant, SCr > 2.26 mg/dL / > 200 µmol/L) — 1 point • **abnormal liver function** (cirrhosis, or bilirubin > 2× ULN with AST/ALT/ALP > 3× ULN) — 1 point	1 or 2
S	**Stroke** (prior)	1
B	**Bleeding** history or predisposition (anemia, thrombocytopenia)	1
L	**Labile INR** (unstable/high INRs, **TTR < 60%**) — only if on warfarin	1
E	**Elderly** — age **> 65**	1
D	**Drugs** (antiplatelets, NSAIDs) — 1 point • **alcohol** (≥ 8 drinks/week) — 1 point	1 or 2
	Maximum	**9**

- **0 = low • 1–2 = moderate • ≥ 3 = HIGH bleeding risk** (~3.7–8.7 major bleeds/100 patient-years).
- **HAS-BLED is NOT used to withhold anticoagulation** — high score → **correct modifiable risk factors** (control BP, stop unnecessary aspirin/NSAIDs, reduce alcohol, improve INR control or switch to DOAC, treat anemia) and **closer follow-up**.
- **Modifiable**: uncontrolled HTN, labile INR, drugs, alcohol • **non-modifiable**: age, prior stroke, prior bleeding, renal/liver disease.
- Other bleeding scores: **ORBIT** (older age, reduced Hb, bleeding history, insufficient kidney function, treatment with antiplatelets), ATRIA, HEMORR2HAGES, **VTE-BLEED**, **IMPROVE-Bleed** (medical inpatients; ≥ 7 = high).

3. Bleeding-Risk Scores After PCI (DAPT decisions)

Score	Components	Interpretation
PRECISE-DAPT	Hemoglobin, WBC, age, CrCl, prior spontaneous bleeding	**≥ 25 = high bleeding risk** → short DAPT (1–3 or 3–6 months)
DAPT score	Age ≥ 75 (−2), 65–74 (−1), < 65 (0) • smoking (1) • diabetes (1) • MI at presentation (1) • prior PCI/MI (1) • paclitaxel stent (1) • stent diameter < 3 mm (1) • HF or LVEF < 30% (2) • vein graft PCI (2)	**≥ 2** → benefit of **prolonged DAPT** (> 12 months) outweighs bleeding • **< 2** → stop at 12 months
ARC-HBR (Academic Research Consortium High Bleeding Risk)	**Major**: long-term OAC, CrCl < 30, Hb < 11, major bleed within 6 months, platelets < 100,000, cirrhosis with portal HTN, active cancer, prior ICH, recent major surgery/trauma • **minor**: age ≥ 75, CrCl 30–59, Hb 11–12.9 (men)/11–11.9 (women), chronic NSAIDs/steroids, prior ischemic stroke	**HBR = ≥ 1 major or ≥ 2 minor** → shortened DAPT
CRUSADE	Hematocrit, CrCl, HR, sex, HF, vascular disease, diabetes, SBP	In-hospital major bleeding in NSTEMI (> 40 = high)

4. Acute Coronary Syndrome and Heart Failure Scores

TIMI risk score — UA/NSTEMI (1 point each)

Item	Item
Age ≥ 65	**Aspirin use in past 7 days**
≥ 3 CAD risk factors (family history, HTN, dyslipidemia, DM, smoking)	**Severe angina** (≥ 2 episodes in 24 h)
Known CAD (stenosis ≥ 50%)	**ST deviation ≥ 0.5 mm**
Positive cardiac biomarker (troponin)	—

- **0–2 = low • 3–4 = intermediate • 5–7 = high** risk (death/MI/urgent revascularization at 14 days) → higher scores favor early invasive strategy.

Other ACS scores

- **GRACE** (in-hospital/6-month mortality): **age, HR, SBP, creatinine, Killip class, cardiac arrest at admission, ST deviation, elevated biomarkers • > 140 = high risk → early invasive (< 24 h)**.
- **HEART score** (ED chest pain): **H**istory, **E**CG, **A**ge, **R**isk factors, **T**roponin — each 0–2 • **0–3 low** (early discharge), **4–6 moderate** (observe/test), **7–10 high** (early invasive).
- **Killip classification** (HF in acute MI): **I** no HF • **II** rales, S3, JVD • **III** acute pulmonary edema • **IV** cardiogenic shock (mortality ↑ with class).
- **TIMI STEMI score** (age, DM/HTN/angina, SBP < 100, HR > 100, Killip II–IV, weight < 67 kg, anterior STE/LBBB, time to treatment > 4 h).

Heart failure and angina classifications

Scale	Classes
NYHA	**I** no limitation • **II** slight limitation (symptoms with ordinary activity) • **III** marked limitation (less than ordinary activity) • **IV** symptoms at rest
ACC/AHA HF stages	**A** at risk • **B** pre-HF (structural disease, no symptoms) • **C** symptomatic • **D** advanced (never go backward)
Forrester / hemodynamic profiles	Warm-dry (A) • **warm-wet (B — most common: diuretics ± vasodilators)** • cold-wet (C: inotropes) • cold-dry (L)
CCS angina	I strenuous exertion only • II slight limitation • III marked limitation (1–2 blocks) • IV at rest/any activity
LVEF categories	HFrEF ≤ 40% • HFmrEF 41–49% • HFpEF ≥ 50% • HFimpEF (improved from ≤ 40% to > 40%)

ASCVD risk estimation

- **Pooled Cohort Equations** (10-year ASCVD risk, age 40–79): < 5% low • 5–7.4% borderline • 7.5–19.9% intermediate • ≥ 20% high.
- **PREVENT equations** (AHA 2023; ages 30–79; total CVD incl. HF; uses eGFR, no race) — used in the 2025 HTN guideline (≥ 7.5% → treat stage 1 HTN).
- **Coronary artery calcium (CAC)**: 0 → may withhold statin • 1–99 → favors statin • ≥ 100 or ≥ 75th percentile → statin.
- **Dutch Lipid Clinic Network score** (familial hypercholesterolemia): **> 8 definite**, 6–8 probable, 3–5 possible (LDL level, family history, xanthomas, corneal arcus, premature CAD) • Simon Broome criteria.

5. Venous Thromboembolism Scores

Wells score — DVT

Criterion	Points
Active cancer • paralysis/paresis/recent cast • bedridden ≥ 3 days or major surgery within 12 weeks • localized tenderness along deep veins • entire leg swollen • calf swelling > 3 cm • pitting edema (symptomatic leg) • collateral superficial veins • previously documented DVT	+1 each
Alternative diagnosis at least as likely	**−2**

- **≤ 1 = DVT unlikely** → D-dimer (negative = excluded) • **≥ 2 = likely** → compression ultrasound.

Wells score — PE

Criterion	Points
Clinical signs/symptoms of DVT	**3**
PE is the most likely diagnosis	**3**
Heart rate > 100	1.5
Immobilization ≥ 3 days or surgery in previous 4 weeks	1.5
Previous DVT/PE	1.5
Hemoptysis	1
Malignancy	1

- **≤ 4 = PE unlikely** → D-dimer (age-adjusted: age × 10 if > 50) • **> 4 = PE likely** → **CTPA** • (three-tier: < 2 low, 2–6 moderate, > 6 high).
- **PERC rule** (only in **low pretest probability**; all 8 must be negative → no testing): **age < 50**, **HR < 100**, **SaO2 ≥ 95%**, no hemoptysis, no estrogen, no prior VTE, no unilateral leg swelling, no surgery/trauma requiring hospitalization in 4 weeks.
- **Revised Geneva score** (objective alternative to Wells) • **YEARS algorithm** (DVT signs, hemoptysis, PE most likely + D-dimer < 1,000 if 0 items, < 500 if ≥ 1).

PE severity — PESI and sPESI

sPESI item (1 point each)	Interpretation
Age > 80 • **cancer** • **chronic cardiopulmonary disease** (HF or chronic lung disease) • **HR ≥ 110** • **SBP < 100** • **SaO2 < 90%**	**0 = low risk** (30-day mortality ~1%) → **outpatient** DOAC • **≥ 1 = higher risk** → admit, assess RV and troponin

- **PESI** (11 variables) classes: **I ≤ 65**, **II 66–85** (low risk) • III 86–105, IV 106–125, **V > 125** (high).
- **Hestia criteria** (outpatient PE if all "no"): hemodynamic instability, thrombolysis/embolectomy needed, active bleeding, O2 need, PE on anticoagulation, IV analgesia, social/medical reasons, CrCl < 30, severe liver disease, pregnancy, HIT history.
- **ESC risk:** high (shock) → thrombolysis • intermediate-high (RV dysfunction + ↑ troponin) → anticoagulate + monitor • intermediate-low • low (sPESI 0).

VTE prophylaxis scores

Score	Items (points)	Action
Padua (medical inpatients)	**Active cancer 3**, **previous VTE 3**, **reduced mobility 3**, **thrombophilia 3**, recent trauma/surgery ≤ 1 month 2, age ≥ 70 1, heart/respiratory failure 1, acute MI/stroke 1, acute infection/rheumatologic disorder 1, BMI ≥ 30 1, hormonal therapy 1	**≥ 4 = high risk** → pharmacologic prophylaxis (if bleeding risk acceptable)
IMPROVE-VTE	Prior VTE 3, thrombophilia 2, lower-limb paralysis 2, cancer 2, immobilization ≥ 7 days 1, ICU/CCU 1, age > 60 1	**≥ 2** → prophylaxis • (IMPROVE-Bleed ≥ 7 = high bleeding risk → mechanical)
Caprini (surgical)	**1 pt**: age 41–60, minor surgery, BMI > 25, swollen legs, varicose veins, pregnancy/postpartum, OC/HRT, sepsis, COPD, MI, HF • **2 pts**: age 61–74, major surgery > 45 min, laparoscopic > 45 min, cancer, bed > 72 h, cast, central line • **3 pts**: age ≥ 75, history of VTE, family history, thrombophilia, HIT • **5 pts**: stroke < 1 month, elective arthroplasty, hip/pelvis/leg fracture, acute spinal cord injury	**0** very low (ambulation) • **1–2** low (IPC) • **3–4** moderate (LMWH/UFH or IPC) • **≥ 5** high (LMWH/UFH + IPC; extended for cancer surgery)
Khorana (cancer outpatients)	Very high-risk site (**stomach, pancreas**) 2 • high-risk site (lung, lymphoma, gynecologic, bladder, testicular) 1 • **platelets ≥ 350,000** 1 • **Hb < 10 or ESA use** 1 • **WBC > 11,000** 1 • **BMI ≥ 35** 1	**≥ 2** → consider **apixaban 2.5 mg BID** or **rivaroxaban 10 mg daily** (or LMWH)

4T score — heparin-induced thrombocytopenia

Category	2 points	1 point	0 points
Thrombocytopenia	Fall > 50% **and** nadir ≥ 20,000	Fall 30–50% or nadir 10,000–19,000	Fall < 30% or nadir < 10,000
Timing of fall	**Day 5–10**, or ≤ 1 day with heparin in past 30 days	Consistent but unclear, > day 10, or ≤ 1 day with exposure 30–100 days ago	< 4 days without recent exposure
Thrombosis	New proven thrombosis, skin necrosis, acute systemic reaction after bolus	Progressive/recurrent or suspected thrombosis, erythematous skin lesions	None
oTher causes	None apparent	Possible	Definite

- **0–3 = low probability** (HIT unlikely — no testing, continue heparin) • **4–5 = intermediate** • **6–8 = high** → stop all heparin, start non-heparin anticoagulant, send PF4 ELISA ± serotonin release assay.

PART B — DIRECT ORAL ANTICOAGULANTS (DOACs) — COMPLETE GUIDE

6. DOAC Pharmacology

Property	Dabigatran	Rivaroxaban	Apixaban	Edoxaban
Target	**Direct thrombin (IIa)** inhibitor	Factor **Xa**	Factor **Xa**	Factor **Xa**
Prodrug	**Yes** (dabigatran etexilate → esterases)	No	No	No
Bioavailability	**3–7%** (capsule must stay intact)	**66% fasting → ~100% with food** (15/20 mg)	~50%	~62%
Food	With or without (food ↓ GI upset)	**15 and 20 mg WITH food** (2.5/10 mg any time)	With or without	With or without
Half-life	12–17 h (↑ in CKD)	5–9 h (11–13 h elderly)	~12 h	10–14 h

Property	Dabigatran	Rivaroxaban	Apixaban	Edoxaban
Renal elimination	**~80%** (most renal)	~33% (unchanged)	**~27%** (least renal)	~50%
Metabolism	Not CYP	**CYP3A4/2J2 + P-gp**	**CYP3A4 + P-gp**	Minimal CYP; **P-gp**
Dialyzable	**Yes** (~60%)	No (protein bound)	No	No
Dosing frequency	BID	Daily (BID first 21 days for VTE)	BID	Daily
Specific reversal	**Idarucizumab**	Andexanet alfa	Andexanet alfa	(Andexanet off-label)
Key ADR	**Dyspepsia/GI upset**, GI bleeding	GI bleeding	**Least GI bleeding**	GI bleeding
Crushable / NG tube	**No** (don't open capsules — ↑ absorption 75%)	Yes (with food after)	Yes	Yes

- **Rapid onset** (peak 1–4 h — no bridging needed when starting) • **short half-life** → missed doses quickly lose protection • **no routine monitoring** • fixed dosing • fewer food/drug interactions than warfarin • **less intracranial hemorrhage** than warfarin.
- **Boxed warnings (all DOACs):** (1) **premature discontinuation ↑ thrombotic events** (stroke) • (2) **spinal/epidural hematoma** with neuraxial anesthesia or spinal puncture • edoxaban: **↓ efficacy in AF if CrCl > 95**.
- **Contraindications / avoid: mechanical heart valves** (dabigatran ↑ thrombosis and bleeding — RE-ALIGN), **moderate–severe mitral stenosis** (AF), active pathologic bleeding, **triple-positive antiphospholipid syndrome**, pregnancy and breastfeeding, severe hepatic impairment (Child-Pugh C — all; rivaroxaban and edoxaban avoid Child-Pugh B), dabigatran/edoxaban with CrCl < 15.

7. DOAC Doses by Indication

Indication	Dabigatran	Rivaroxaban	Apixaban	Edoxaban
Nonvalvular AF (stroke prevention)	**150 mg BID**	**20 mg daily with evening meal**	**5 mg BID**	**60 mg daily**
AF — dose reduction	**75 mg BID if CrCl 15–30** • 75 mg BID if CrCl 30–50 + dronedarone or ketoconazole • avoid CrCl < 15	**15 mg daily if CrCl 15–50** (avoid < 15)	**2.5 mg BID if ≥ 2 of 3: age ≥ 80, weight ≤ 60 kg, SCr ≥ 1.5 mg/dL**	**30 mg if CrCl 15–50 • don't use if CrCl > 95 •** avoid < 15
VTE treatment — initiation	**Parenteral 5–10 days first**	**15 mg BID × 21 days** (with food)	**10 mg BID × 7 days**	**Parenteral 5–10 days first**
VTE treatment — maintenance	**150 mg BID**	**20 mg daily** (with food)	**5 mg BID**	**60 mg daily** (30 mg if CrCl 15–50, ≤ 60 kg, or P-gp inhibitors)
VTE — extended (after ≥ 6 months)	150 mg BID	**10 mg daily** (20 mg if high risk)	**2.5 mg BID**	60 mg daily
VTE prophylaxis — hip replacement	110 mg day 1 → 220 mg daily × 28–35 d	**10 mg daily × 35 days**	**2.5 mg BID × 35 days**	(not US-approved)
VTE prophylaxis — knee replacement	(not US-approved)	**10 mg daily × 12 days**	**2.5 mg BID × 12 days**	—
Medically ill inpatients	—	10 mg daily × 31–39 days (selected)	—	—
CAD / PAD (with aspirin)	—	**2.5 mg BID + aspirin 81 mg** (COMPASS, VOYAGER)	—	—
Cancer-associated VTE outpatient prophylaxis (Khorana ≥ 2)	—	10 mg daily	2.5 mg BID	—

Indication	Dabigatran	Rivaroxaban	Apixaban	Edoxaban
Pediatric VTE	Approved (weight-based pellets/capsules)	Approved (suspension)	Limited	—

- **VTE renal limits:** dabigatran avoid CrCl < 30 • rivaroxaban avoid < 15 (caution 15–30) • apixaban no adjustment (limited data < 25; used in ESRD) • edoxaban 30 mg CrCl 15–50, avoid < 15.
- **Calculate CrCl with Cockcroft–Gault using actual body weight** (as in pivotal trials) • reassess renal function at least **yearly**; every **CrCl/10 months** if CrCl < 60 (e.g., CrCl 40 → every 4 months) or more often with acute illness.
- **AF + PCI:** apixaban 5 mg BID, rivaroxaban 15 mg daily (10 mg if CrCl 30–50), dabigatran 110 or 150 mg BID, edoxaban 60 mg + clopidogrel • aspirin ≤ 1 week (≤ 30 days).
- **Don't under-dose** (off-label low doses ↑ stroke) — use label criteria exactly.

8. DOAC Drug Interactions

Interaction	Effect	Management
Combined strong CYP3A4 + P-gp inhibitors: ketoconazole, itraconazole, posaconazole, voriconazole, **ritonavir/cobicistat**, clarithromycin (rivaroxaban caution)	↑ rivaroxaban and apixaban levels → bleeding	**Rivaroxaban: avoid • apixaban: ↓ dose by 50%** (5 → 2.5 mg BID; if already 2.5 mg → avoid)
Combined strong CYP3A4 + P-gp inducers: **rifampin**, **carbamazepine**, **phenytoin**, phenobarbital, **St John's wort**	↓ levels of all DOACs → thrombosis	**Avoid with all DOACs** (use warfarin or LMWH)
P-gp inhibitors: **dronedarone**, **ketoconazole**, verapamil, amiodarone, quinidine, clarithromycin, cyclosporine, ticagrelor	↑ **dabigatran** and **edoxaban**	Dabigatran AF: 75 mg BID if CrCl 30–50 + dronedarone/ketoconazole; avoid if CrCl < 30 • edoxaban VTE: 30 mg with verapamil, quinidine, azithro/clarithro/erythromycin, oral itra/ketoconazole
Antiplatelets (aspirin, clopidogrel, prasugrel, ticagrelor), **NSAIDs**	Additive bleeding	Avoid unless indicated (ACS/PCI) • PPI for GI protection • acetaminophen for pain
SSRIs/SNRIs	↓ platelet serotonin → bleeding	Monitor
Other anticoagulants	Additive bleeding	Never combine (except during switching)
Herbals: ginkgo, garlic, ginseng, fish oil (high dose), turmeric	↑ bleeding	Avoid
Antacids/PPIs	Slight ↓ dabigatran absorption	Not clinically significant usually

9. Switching Between Anticoagulants

From → To	How
Warfarin → DOAC	Stop warfarin, start DOAC when INR is: **apixaban < 2.0 • dabigatran < 2.0 • rivaroxaban < 3.0 • edoxaban ≤ 2.5**
DOAC → warfarin	**Apixaban/rivaroxaban**: start warfarin, continue DOAC until INR ≥ 2 (draw INR just before next DOAC dose — DOACs ↑ INR) or overlap with parenteral • **edoxaban**: halve edoxaban dose (60 → 30 mg) and start warfarin; stop edoxaban when INR ≥ 2 • **dabigatran**: start warfarin **3 days** before stopping if CrCl ≥ 50 (**2 days** if CrCl 30–50, **1 day** if 15–30)
LMWH → DOAC	Start DOAC **0–2 h before the next scheduled LMWH dose** (stop LMWH)
UFH infusion → DOAC	Start DOAC **at the time the infusion is stopped** (dabigatran: when stopped)
DOAC → parenteral (LMWH/UFH)	Start parenteral when the **next DOAC dose would be due**

From → To	How
DOAC → another DOAC	Start the new DOAC when the next dose of the old one would be due

10. Periprocedural Management (PAUSE approach)

DOAC	Low/moderate bleeding-risk procedure — last dose	High bleeding-risk procedure — last dose
Apixaban, rivaroxaban, edoxaban	**Hold 1 day** before (skip 2 doses of BID / 1 dose of daily)	**Hold 2 days** before
Dabigatran — CrCl ≥ 50	**Hold 1 day**	**Hold 2 days**
Dabigatran — CrCl 30–50	**Hold 2 days**	**Hold 4 days**

- **Resume** 24 h after low-bleeding-risk procedures and **48–72 h** after high-risk procedures (when hemostasis secured) • full effect within hours (no bridging).
- **No heparin bridging for DOACs** • minimal-risk procedures (dental cleaning/extraction, cataract, minor skin) → may continue or skip one dose.
- **Neuraxial anesthesia** (ASRA): hold apixaban/rivaroxaban/edoxaban **72 h**, dabigatran **72–120 h** (by CrCl) before catheter placement • remove catheter ≥ 6 h before the next dose (avoid DOAC with indwelling catheter).
- **Warfarin**: hold 5 days, check INR day before • bridging with LMWH only for high thrombotic risk (mechanical mitral valve, recent stroke/VTE < 3 months) — BRIDGE trial.

11. Bleeding Management and Reversal

Agent	Reverses	Dose / notes
Idarucizumab (Praxbind)	**Dabigatran**	**5 g IV** (two 2.5 g vials back-to-back) • humanized antibody fragment • immediate • thrombotic risk after reversal • dialysis also removes dabigatran
Andexanet alfa (Andexxa)	**Apixaban, rivaroxaban** (edoxaban off-label)	Recombinant decoy factor Xa • **low dose**: 400 mg bolus (30 mg/min) + 4 mg/min × 120 min • **high dose**: 800 mg bolus + 8 mg/min × 120 min — high dose if last apixaban dose > 5 mg or rivaroxaban > 10 mg (or unknown) within 8 h • **boxed: thromboembolism, MI, stroke, cardiac arrest** • availability limited/expensive
4-factor PCC (Kcentra)	Xa inhibitors (off-label), warfarin	**50 units/kg** (max 5,000) for Xa inhibitors (or fixed 2,000 units) • warfarin: 25–50 units/kg by INR + vitamin K
Activated PCC (FEIBA)	Dabigatran (if idarucizumab unavailable)	50 units/kg
Activated charcoal	Any DOAC	If ingested within 2–4 h
Hemodialysis	**Dabigatran only**	Not effective for Xa inhibitors (highly protein bound)
Protamine	UFH (full), LMWH (~60%)	Not for DOACs or fondaparinux
Vitamin K	Warfarin only	No effect on DOACs

- **General bleeding management:** hold DOAC, local hemostasis, fluids/transfusion, identify last dose time and renal function (half-life), tranexamic acid for some bleeds • minor bleeding → hold 1 dose • life-threatening (ICH, hemodynamic instability) → specific reversal.
- **Lab tests:** PT/INR and aPTT are **not reliable** for DOACs (a normal aPTT doesn't exclude dabigatran; normal PT doesn't exclude apixaban) • **drug-calibrated anti-Xa** (Xa inhibitors) • **dilute thrombin time / ecarin clotting time** (dabigatran) • normal thrombin time excludes dabigatran.

12. DOACs in Special Populations

Population	Recommendation
Severe renal impairment / dialysis	**Apixaban** preferred (5 mg BID; 2.5 if ≥ 2 criteria) or warfarin • avoid dabigatran (CrCl < 15–30), edoxaban/rivaroxaban < 15
Hepatic impairment	Child-Pugh A: all OK • **Child-Pugh B**: apixaban/dabigatran with caution; **avoid rivaroxaban and edoxaban** • **Child-Pugh C: avoid all**
Obesity (BMI > 40 or > 120 kg)	**Apixaban or rivaroxaban** standard doses acceptable (ISTH 2021) • avoid dabigatran and edoxaban • consider drug-specific levels
Bariatric surgery	Avoid DOACs initially (absorption uncertain; esp. rivaroxaban/dabigatran) → parenteral or warfarin, then consider DOAC with levels
Elderly / low body weight	Use DOAC (less ICH than warfarin) at label doses • apixaban dose criteria • falls are not a reason to withhold
Cancer	VTE: apixaban, rivaroxaban, edoxaban (or LMWH — especially GI/GU cancers, DDIs with chemotherapy)
Antiphospholipid syndrome	**Warfarin** (INR 2–3) — avoid DOACs (especially triple-positive, arterial events)
Mechanical heart valves	**Warfarin only** — DOACs contraindicated
Bioprosthetic valves / TAVR, mitral regurgitation	DOACs acceptable (after 3 months for surgical bioprosthetic)
Moderate–severe mitral stenosis + AF	**Warfarin**
Pregnancy / breastfeeding	**Avoid DOACs** → LMWH (pregnancy); LMWH or warfarin (breastfeeding)
LV thrombus	Warfarin or DOAC (DOACs increasingly used)
Heparin-induced thrombocytopenia	DOACs acceptable (stable patients without life-threatening thrombosis)
Children	Dabigatran and rivaroxaban have pediatric approvals (VTE)

DOAC patient counseling

- Take at the **same time daily** • **rivaroxaban 15/20 mg with food** (evening meal for AF) • **dabigatran: swallow whole, don't open capsules, keep in original bottle** (use within 4 months of opening; blister packs protect from moisture) • apixaban/rivaroxaban can be crushed.
- **Missed dose:** BID drugs (apixaban, dabigatran) — take if within 6 h of missed time, else skip • once-daily (rivaroxaban, edoxaban) — take the same day, don't double • **rivaroxaban 15 mg BID phase** — take immediately even if 2 tablets at once (30 mg/day).
- **Don't stop without talking to provider** (boxed — stroke) • inform dentists/surgeons • avoid NSAIDs/aspirin/herbals • report bleeding signs (black stools, blood in urine, vomiting blood, severe headache, unusual bruising) • carry anticoagulant card • monitoring: CBC, SCr, LFTs at baseline and at least yearly.

DOAC vs warfarin vs LMWH — quick comparison

Feature	DOACs	Warfarin	LMWH
Onset	Rapid (1–4 h)	Slow (5–7 days) → needs overlap	Rapid (SC)
Monitoring	None routine	**INR**	Anti-Xa (selected)
Food/DDIs	Few (P-gp/3A4)	**Many** (vitamin K, CYP2C9)	Few
Renal	Dose adjust	No adjustment	Adjust (CrCl < 30)
Reversal	Idarucizumab / andexanet / PCC	Vitamin K + 4F-PCC	Protamine (partial)
Best for	Most AF and VTE	Mechanical valves, MS, APS, ESRD, cost	Pregnancy, cancer (GI/GU), bridging

PART C — NEUROLOGY AND STROKE SCORES

Score	Components	Interpretation
Glasgow Coma Scale (GCS)	**Eye** 1–4 (spontaneous 4, to voice 3, to pain 2, none 1) • **Verbal** 1–5 (oriented 5, confused 4, inappropriate words 3, sounds 2, none 1) • **Motor** 1–6 (obeys 6, localizes 5, withdraws 4, flexion 3, extension 2, none 1)	**3–15** • **≤ 8 → intubate** ("8 intubate") • 9–12 moderate • 13–15 mild
NIHSS	Consciousness, gaze, visual fields, facial palsy, motor arms/legs, ataxia, sensory, language, dysarthria, neglect	**0–42** • 0 none • 1–4 minor • 5–15 moderate • 16–20 moderate–severe • > 20 severe • guides thrombolysis/thrombectomy
ABCD2 (stroke risk after TIA)	**A**ge ≥ 60 (1) • **B**P ≥ 140/90 (1) • **C**linical: unilateral weakness (2), speech disturbance without weakness (1) • **D**uration ≥ 60 min (2), 10–59 min (1) • **D**iabetes (1)	**0–3 low** • **4–5 moderate** • **6–7 high** → DAPT (aspirin + clopidogrel × 21 days) for high-risk TIA/minor stroke
Modified Rankin Scale (mRS)	Disability	0 no symptoms → 5 severe disability → 6 death
ICH score	GCS 3–4 (2), 5–12 (1) • volume ≥ 30 mL (1) • intraventricular hemorrhage (1) • infratentorial (1) • age ≥ 80 (1)	0–6 • 30-day mortality ↑ with score
BE-FAST	**B**alance, **E**yes, **F**ace, **A**rm, **S**peech, **T**ime	Stroke recognition
Hoehn and Yahr (Parkinson's)	Stage 1 unilateral → 2 bilateral → 3 postural instability → 4 severe disability → 5 wheelchair/bedbound	Staging
EDSS (multiple sclerosis)	0 (normal) → 10 (death)	Disability; ≥ 6 needs walking aid
MMSE / MoCA (cognition)	MMSE 0–30 • MoCA 0–30	MMSE ≥ 24 normal; 20–24 mild; 13–20 moderate; < 12 severe dementia • **MoCA < 26 abnormal** (more sensitive for MCI)
Migraine: MIDAS	Disability days in 3 months	I (0–5) little → IV (≥ 21) severe → preventive therapy

PART D — INFECTION, SEPSIS AND ICU SCORES

Score	Components	Interpretation
CURB-65 (CAP)	**C**onfusion • **U**rea (BUN) > 19 mg/dL (> 7 mmol/L) • **R**R ≥ 30 • **B**P: SBP < 90 or DBP ≤ 60 • age ≥ **65** (1 each)	**0–1 outpatient** • **2 admit (ward)** • **≥ 3 severe → consider ICU** (CRB-65 without BUN)
PSI / PORT (CAP)	20 variables (age, sex, comorbidities, exam, labs)	Classes I–II outpatient • III short stay/observation • IV–V inpatient (IDSA prefers PSI)
IDSA/ATS severe CAP	Major: septic shock on vasopressors, mechanical ventilation • Minor (≥ 3): RR ≥ 30, P/F ≤ 250, multilobar, confusion, BUN ≥ 20, WBC < 4,000, platelets < 100,000, T < 36 °C, hypotension needing fluids	1 major or ≥ 3 minor → ICU
qSOFA	**RR ≥ 22** • **altered mentation** • **SBP ≤ 100**	≥ 2 → high risk of poor outcome • not a stand-alone sepsis screen (SSC 2021)
SOFA	Respiration (P/F), coagulation (platelets), liver (bilirubin), cardiovascular (MAP/vasopressors), CNS (GCS), renal (SCr/UO) — 0–4 each	**↑ ≥ 2 = organ dysfunction → sepsis** (Sepsis-3)

Score	Components	Interpretation
SIRS	T > 38 or < 36 • HR > 90 • RR > 20 or PaCO2 < 32 • WBC > 12,000 or < 4,000 or > 10% bands	≥ 2 = SIRS (sensitive, not specific)
NEWS2	RR, SpO2, O2 use, SBP, HR, consciousness, temperature	≥ 5 urgent review • ≥ 7 emergency response
Centor / McIsaac (GAS pharyngitis)	Fever > 38 • tonsillar exudates/swelling • tender anterior cervical nodes • absence of cough (1 each) • age 3–14 (+1), ≥ 45 (−1)	**0–1**: no test/no antibiotic • **2–3**: rapid antigen test → treat if positive • **≥ 4**: test (or empiric in some guidelines)
Modified Duke criteria (endocarditis)	Major: positive blood cultures (typical organism ×2), imaging evidence • minor: predisposition, fever, vascular, immunologic, microbiologic	Definite: 2 major, 1 major + 3 minor, or 5 minor
LRINEC (necrotizing fasciitis)	CRP, WBC, Hb, Na, SCr, glucose	≥ 6 suspicious, ≥ 8 strongly predictive (don't delay surgery)
Anthonisen criteria (COPD exacerbation)	↑ dyspnea, ↑ sputum volume, ↑ sputum purulence	Type I (all 3) or type II (2 incl. purulence) → antibiotics
C. difficile severity	WBC, SCr, hypotension/ileus/megacolon	Non-severe (WBC ≤ 15,000 + SCr < 1.5) • severe (WBC > 15,000 or SCr > 1.5) • fulminant
APACHE II	Age, chronic health, 12 physiologic variables	0–71 • > 25 high mortality • IAI high-severity > 15
Shock index	HR / SBP	> 1.0 suggests shock/hypovolemia

ICU sedation, pain and delirium tools

Tool	Interpretation
RASS (Richmond Agitation-Sedation Scale)	+4 combative → 0 alert and calm → −5 unarousable • **goal −2 to 0** (light sedation)
SAS (Riker)	1 unarousable → 4 calm/cooperative → 7 dangerous agitation • goal 3–4
BPS (Behavioral Pain Scale)	3–12 • **> 5 = significant pain**
CPOT (Critical-Care Pain Observation Tool)	0–8 • **> 2 = pain**
CAM-ICU	(1) acute change/fluctuating + (2) inattention + either (3) altered LOC or (4) disorganized thinking = delirium
ICDSC	0–8 • **≥ 4 = delirium**
CIWA-Ar (alcohol withdrawal)	0–67 • **< 8–10 minimal** • 10–15 moderate • **≥ 15–20 severe** → symptom-triggered benzodiazepines at ≥ 8–10
COWS (opioid withdrawal)	5–12 mild • 13–24 moderate • 25–36 moderately severe • > 36 severe • start buprenorphine when COWS ≥ 8–12 (objective withdrawal)
Pain NRS	0–10 self-report • goal < 4

PART E — GI AND LIVER SCORES

Child-Pugh score (cirrhosis severity)

Parameter	1 point	2 points	3 points
Total bilirubin (mg/dL)	< 2	2–3	> 3
Albumin (g/dL)	> 3.5	2.8–3.5	< 2.8
INR	< 1.7	1.7–2.3	> 2.3
Ascites	None	Mild (controlled with diuretics)	Moderate–severe (refractory)
Hepatic encephalopathy	None	Grade 1–2 (controlled)	Grade 3–4 (refractory)

- **Class A: 5–6** (well compensated) • **Class B: 7–9** (significant compromise) • **Class C: 10–15** (decompensated) → used for **drug dose adjustment** (e.g., avoid rivaroxaban/edoxaban in B; many drugs CI in C) • mnemonic "**ABCDE**" (albumin, bilirubin, coagulation, distension/ascites, encephalopathy).

Score	Components	Interpretation
MELD / MELD-Na (MELD 3.0 adds albumin, sex)	**Bilirubin, INR, creatinine** (+ **sodium**)	6–40 • 3-month mortality • **liver transplant priority** • **≥ 15 → transplant referral** • TIPS risk
West Haven (hepatic encephalopathy)	Grade 0 (minimal — only on testing) • **1** mild confusion, short attention, sleep reversal • **2** lethargy, disorientation, **asterixis** • **3** somnolence, gross confusion • **4** coma	Grade ≥ 2 overt HE → lactulose (+ rifaximin for recurrence)
Maddrey discriminant function (alcoholic hepatitis)	4.6 × (PT − control) + bilirubin	**≥ 32 → prednisolone 40 mg × 28 days** (if no contraindication)
Lille score	Day 7 response to steroids	**≥ 0.45 → non-responder → stop steroids**
MELD in alcoholic hepatitis	—	> 20 severe → consider steroids
FIB-4 (fibrosis, MASLD/HCV)	Age, AST, ALT, platelets	**< 1.3 low** (rule out advanced fibrosis) • 1.3–2.67 indeterminate (→ elastography) • **> 2.67 high**
APRI	AST/ULN ÷ platelets × 100	> 1.0–1.5 suggests significant fibrosis/cirrhosis (> 2)
Glasgow-Blatchford (upper GI bleed — pre-endoscopy)	BUN, Hb, SBP, HR, melena, syncope, hepatic disease, HF	**0–1 → low risk, outpatient management** • higher → admission/endoscopy
Rockall (UGIB — post-endoscopy)	Age, shock, comorbidity, diagnosis, stigmata of recent hemorrhage	Rebleeding and mortality risk • ≤ 2 low
Forrest classification (peptic ulcer bleeding)	**Ia** spurting • **Ib** oozing • **IIa** non-bleeding visible vessel • **IIb** adherent clot • **IIc** flat pigmented spot • **III** clean base	**Ia, Ib, IIa (± IIb) = high risk** → endoscopic therapy + **high-dose IV PPI × 72 h** • IIc, III low risk → oral PPI, early feeding
Los Angeles classification (erosive esophagitis)	**A** mucosal breaks ≤ 5 mm • **B** > 5 mm, not continuous • **C** continuous between folds < 75% • **D** ≥ 75% circumference	C/D severe → long-term PPI • B–D confirms GERD
Ranson criteria (pancreatitis)	Admission: age > 55, WBC > 16,000, glucose > 200, LDH > 350, AST > 250 • 48 h: Hct ↓ > 10%, BUN ↑ > 5, Ca < 8, PaO2 < 60, base deficit > 4, fluid sequestration > 6 L	**≥ 3 = severe** pancreatitis

Score	Components	Interpretation
BISAP (pancreatitis)	**B**UN > 25 • **I**mpaired mental status • **S**IRS • **A**ge > 60 • **P**leural effusion	**≥ 3 = ↑ mortality**
Revised Atlanta (pancreatitis severity)	Organ failure, local complications	**Mild** (none) • **moderately severe** (transient organ failure < 48 h or local complications) • **severe** (persistent organ failure > 48 h)
Truelove and Witts (acute severe UC)	**≥ 6 bloody stools/day** + ≥ 1 systemic sign: HR > 90, T > 37.8 °C, Hb < 10.5, ESR > 30 (CRP > 30)	Acute severe UC → hospitalize, **IV methylprednisolone/hydrocortisone** → rescue infliximab/cyclosporine
Mayo score (UC)	Stool frequency, rectal bleeding, endoscopy, physician global (0–3 each)	0–2 remission • 3–5 mild • 6–10 moderate • 11–12 severe
Montreal classification	Crohn's: Age (A), Location (L1 ileal, L2 colonic, L3 ileocolonic, L4 upper GI), Behavior (B1 inflammatory, B2 stricturing, B3 penetrating, p perianal) • UC: E1 proctitis, E2 left-sided, E3 extensive	Guides therapy (proctitis → rectal mesalamine)
CDAI / Harvey-Bradshaw (Crohn's)	CDAI < 150 remission, 150–220 mild, 220–450 moderate, > 450 severe • HBI < 5 remission	Disease activity

PART F — RENAL, FLUIDS AND ELECTROLYTE CALCULATIONS

Score / formula	Details
KDIGO AKI staging	**Stage 1**: SCr 1.5–1.9× baseline **or ↑ ≥ 0.3 mg/dL within 48 h**; UO < 0.5 mL/kg/h for 6–12 h • **Stage 2**: SCr 2.0–2.9×; UO < 0.5 for ≥ 12 h • **Stage 3**: SCr ≥ 3× or ≥ 4.0 mg/dL or RRT; UO < 0.3 for ≥ 24 h or anuria ≥ 12 h
KDIGO CKD (G and A)	**G1** ≥ 90 • **G2** 60–89 • **G3a** 45–59 • **G3b** 30–44 • **G4** 15–29 • **G5** < 15 • albuminuria **A1** < 30, **A2** 30–300, **A3** > 300 mg/g • CKD = abnormality > 3 months
Cockcroft-Gault	CrCl = [(140 − age) × weight] / (72 × SCr) × 0.85 (female) — **used for drug dosing**
CKD-EPI 2021	Race-free eGFR — CKD staging
FENa	(UNa × PCr) / (PNa × UCr) × 100 • **< 1% prerenal** • **> 2% intrinsic (ATN)** • on diuretics use **FEUrea < 35% = prerenal**
Anion gap	Na − (Cl + HCO3) = 8–12 • albumin-corrected: + 2.5 × (4 − albumin)
Winter's formula	Expected PaCO2 = 1.5 × HCO3 + 8 ± 2
Serum osmolality	2 × Na + glucose/18 + BUN/2.8 (275–295) • osmolal gap > 10 → toxic alcohols
Corrected sodium (hyperglycemia)	Na + 1.6 × (glucose − 100)/100
Corrected calcium	Ca + 0.8 × (4 − albumin)
Free water deficit	TBW × [(Na/140) − 1] • TBW = 0.6 × wt (men), 0.5 (women), 0.5/0.45 elderly
Adrogué–Madias	ΔNa per 1 L infusate = (infusate Na − serum Na) / (TBW + 1) • max correction 8–10 mEq/L/24 h
Bicarbonate deficit	0.5 × weight × (desired − measured HCO3)
Mentzer index (microcytosis)	MCV / RBC count • **< 13 → thalassemia trait** • **> 13 → iron deficiency**
Reticulocyte production index	Retic % × (Hct/45) / maturation factor • > 2 adequate marrow response (hemolysis/bleeding), < 2 hypoproliferative

PART G — PULMONARY SCORES

Score	Interpretation
GOLD spirometric grade (post-bronchodilator FEV1/FVC < 0.70)	**GOLD 1** FEV1 ≥ 80% • **2** 50–79% • **3** 30–49% • **4** < 30%
GOLD groups (2023+)	**A**: 0–1 moderate exacerbation, mMRC 0–1/CAT < 10 → bronchodilator • **B**: 0–1 exacerbation, mMRC ≥ 2/CAT ≥ 10 → **LAMA + LABA** • **E**: ≥ 2 moderate or ≥ 1 hospitalization → **LAMA + LABA** (+ ICS if eosinophils ≥ 300)
mMRC dyspnea scale	0 (strenuous exercise only) → 4 (too breathless to leave house) • **≥ 2 = more symptomatic**
CAT (COPD Assessment Test)	0–40 • **≥ 10 = more symptomatic**
GINA asthma control (past 4 weeks)	Daytime symptoms > 2×/week • night waking • SABA reliever > 2×/week • activity limitation → **0 = well controlled**, **1–2 partly**, **3–4 uncontrolled**
ACT (Asthma Control Test)	5–25 • **≥ 20 well controlled** • 16–19 not well controlled • ≤ 15 very poorly controlled
NAEPP asthma severity (intermittent vs persistent)	Intermittent: symptoms ≤ 2 days/week, night ≤ 2×/month, FEV1 > 80% • persistent mild → moderate → severe (daily/continual symptoms, nightly, FEV1 < 60%)
Asthma exacerbation severity	Mild/moderate (talks in phrases, SpO2 90–95%) • severe (talks in words, RR > 30, HR > 120, SpO2 < 90%, PEF ≤ 50%) • life-threatening (drowsy, silent chest)
BODE index (COPD mortality)	BMI, Obstruction (FEV1), Dyspnea (mMRC), Exercise (6-min walk) — 0–10

PART H — PSYCHIATRY, ENDOCRINE, RHEUMATOLOGY AND NUTRITION

Score	Interpretation
PHQ-9 (depression)	0–27 • **5 mild, 10 moderate, 15 moderately severe, 20 severe** • **≥ 10 = positive screen** • **item 9 (suicidal thoughts)** → immediate safety assessment • PHQ-2 ≥ 3 → full PHQ-9 • response = ≥ 50% ↓, remission < 5
GAD-7 (anxiety)	0–21 • 5 mild, 10 moderate, 15 severe • ≥ 10 positive
HAM-D / MADRS	Clinician-rated depression severity (research/clinical trials)
YMRS (Young Mania Rating Scale)	Mania severity (> 20 moderate–severe)
AUDIT / AUDIT-C (alcohol)	AUDIT ≥ 8 hazardous use • AUDIT-C ≥ 4 men / ≥ 3 women positive
CAGE	**C**ut down, **A**nnoyed, **G**uilty, **E**ye-opener • ≥ 2 = clinically significant
Columbia Suicide Severity Rating Scale	Suicide risk screening
Epworth Sleepiness Scale	> 10 excessive daytime sleepiness
Burch–Wartofsky (thyroid storm)	Temperature, CNS, GI/hepatic, HR, HF, AF, precipitant • **≥ 45 = storm** • 25–44 impending • < 25 unlikely
DKA severity	Mild pH 7.25–7.30 (HCO3 15–18) • moderate 7.00–7.24 (10–14) • severe < 7.00 (< 10) or stupor
FRAX (10-year fracture probability)	Treat osteopenia if **major osteoporotic ≥ 20%** or **hip ≥ 3%**
T-score / Z-score	T ≤ −2.5 osteoporosis • −1 to −2.5 osteopenia • Z ≤ −2.0 below expected (premenopausal, men < 50)
BMI	< 18.5 underweight • 18.5–24.9 normal • 25–29.9 overweight • 30–34.9 class I • 35–39.9 class II • ≥ 40 class III (Asian: overweight ≥ 23, obese ≥ 25–27.5)
DAS28 (RA activity)	**< 2.6 remission** • ≤ 3.2 low • 3.2–5.1 moderate • **> 5.1 high**
CDAI (RA)	≤ 2.8 remission • ≤ 10 low • ≤ 22 moderate • > 22 high

Score	Interpretation
ACR/EULAR 2010 RA classification	Joints (0–5), serology RF/anti-CCP (0–3), acute-phase reactants (0–1), duration ≥ 6 weeks (0–1) → **≥ 6/10 = definite RA**
Kellgren–Lawrence (OA X-ray)	Grade 0 none → 4 severe (large osteophytes, marked narrowing, sclerosis)
ACR/EULAR gout classification	Entry: ≥ 1 episode of swelling/pain • sufficient: MSU crystals
NRS-2002 (nutrition risk)	**≥ 3 = at risk** → nutrition plan
NUTRIC (ICU nutrition risk)	**≥ 5 (with IL-6) / ≥ 6 = high risk** → early aggressive nutrition
MUST	0 low, 1 medium, ≥ 2 high malnutrition risk
Rumack–Matthew nomogram	Acetaminophen level 4–24 h after acute ingestion • treatment line 150 mcg/mL at 4 h → NAC
Apgar (newborn)	Appearance, Pulse, Grimace, Activity, Respiration (0–2 each) • 7–10 normal

Exam Traps and Frequently Tested Points

1. **CHA2DS2-VASc:** age ≥ 75 and stroke/TIA = **2 points** each • OAC if **≥ 2 men / ≥ 3 women** • female sex alone not an indication.
2. **HAS-BLED ≥ 3 = high bleeding risk** → fix modifiable factors (BP, NSAIDs/aspirin, alcohol, labile INR) — **never a reason by itself to withhold OAC**.
3. **HAS-BLED letters:** HTN (SBP > 160), Abnormal renal/liver (1 each), Stroke, Bleeding, Labile INR, Elderly > 65, Drugs/alcohol (1 each) • max 9.
4. **TIMI UA/NSTEMI**: 7 items (age ≥ 65, ≥ 3 risk factors, known CAD, aspirin in 7 days, ≥ 2 anginal episodes/24 h, ST deviation, + biomarker) • ≥ 5 high.
5. **GRACE > 140 → early invasive < 24 h** • HEART 0–3 discharge • Killip IV = cardiogenic shock • DAPT score ≥ 2 → prolonged DAPT • PRECISE-DAPT ≥ 25 → short DAPT.
6. **Wells DVT ≤ 1 / PE ≤ 4 → D-dimer** • PERC only in low probability • sPESI 0 → outpatient • Padua ≥ 4 → prophylaxis • Caprini ≥ 5 high • Khorana ≥ 2 → apixaban/rivaroxaban.
7. **4T score:** 0–3 low (HIT unlikely) • 4–5 intermediate • 6–8 high • timing 5–10 days.
8. **Dabigatran:** direct thrombin inhibitor, prodrug, **80% renal**, dialyzable, dyspepsia, don't open capsules, original bottle, **idarucizumab 5 g**.
9. **Apixaban:** least renal (27%), least GI bleeding, OK in ESRD • AF 5 mg BID (2.5 if ≥ 2 of: ≥ 80 y, ≤ 60 kg, SCr ≥ 1.5) • VTE 10 mg BID × 7 d → 5 mg BID → 2.5 mg BID extended.
10. **Rivaroxaban:** 15/20 mg **with food** • AF 20 mg (15 if CrCl 15–50) • VTE 15 mg BID × 21 d → 20 mg → 10 mg extended • CAD/PAD 2.5 mg BID + aspirin.
11. **Edoxaban:** 60 mg (30 if CrCl 15–50, ≤ 60 kg, P-gp inhibitors) • **not in AF if CrCl > 95** • needs 5–10 days parenteral for VTE (like dabigatran).
12. **DOACs contraindicated** with mechanical valves, moderate–severe mitral stenosis (AF), triple-positive APS, pregnancy • avoid with rifampin, carbamazepine, phenytoin, St John's wort.
13. **Strong 3A4 + P-gp inhibitors:** avoid rivaroxaban; apixaban ↓ 50% • P-gp inhibitors ↑ dabigatran/edoxaban (dronedarone, ketoconazole, verapamil).
14. **Switching warfarin → DOAC:** start when INR < 2 (apixaban, dabigatran), < 3 (rivaroxaban), ≤ 2.5 (edoxaban) • LMWH → DOAC: 0–2 h before next LMWH dose.
15. **Periprocedural:** no bridging for DOACs • Xa inhibitors hold 1 day (low risk) / 2 days (high risk) • dabigatran longer if CrCl < 50 • resume 24 h / 48–72 h.
16. **Reversal:** dabigatran → idarucizumab (or dialysis) • apixaban/rivaroxaban → andexanet alfa or 4F-PCC 50 units/kg • vitamin K and protamine don't reverse DOACs.
17. **PT/aPTT unreliable for DOACs** → drug-calibrated anti-Xa / dilute thrombin time • DOAC CrCl by Cockcroft–Gault (actual body weight), recheck every CrCl/10 months if < 60.
18. **GCS ≤ 8 → intubate** • NIHSS 0–42 • ABCD2 ≥ 4 → DAPT 21 days after TIA • MMSE < 24 / MoCA < 26 abnormal.
19. **CURB-65:** 0–1 home, 2 admit, ≥ 3 ICU • qSOFA ≥ 2 (RR ≥ 22, AMS, SBP ≤ 100) • SOFA ↑ ≥ 2 = sepsis • Centor ≥ 2–3 → test.
20. **RASS goal −2 to 0** • BPS > 5 / CPOT > 2 = pain • ICDSC ≥ 4 delirium • CIWA ≥ 8–10 treat • COWS ≥ 8–12 start buprenorphine.

21. **Child-Pugh A 5–6, B 7–9, C 10–15** • MELD ≥ 15 transplant referral • West Haven grade 2 = asterixis • Maddrey ≥ 32 → prednisolone • Lille ≥ 0.45 stop steroids.
22. **Glasgow-Blatchford 0–1 → outpatient** • **Forrest Ia/Ib/IIa = high risk** → endoscopic therapy + IV PPI 72 h • LA grade C/D severe esophagitis.
23. **Ranson ≥ 3 / BISAP ≥ 3 = severe pancreatitis** • revised Atlanta severe = organ failure > 48 h • Truelove–Witts: ≥ 6 bloody stools + systemic sign → IV steroids.
24. **KDIGO AKI stage 1:** SCr ↑ ≥ 0.3 in 48 h or 1.5–1.9× • stage 3: ≥ 3× or ≥ 4.0 or RRT • FENa < 1% prerenal (FEUrea < 35% on diuretics) • Mentzer < 13 thalassemia.
25. **GOLD 1–4 by FEV1** (80/50/30) • groups A, B (LAMA+LABA), E (LAMA+LABA ± ICS if eos ≥ 300) • mMRC ≥ 2 / CAT ≥ 10 symptomatic • ACT ≥ 20 controlled.
26. **PHQ-9 ≥ 10 positive** (item 9 = suicide) • GAD-7 ≥ 10 • AUDIT ≥ 8 • CAGE ≥ 2 • Burch–Wartofsky ≥ 45 storm • DAS28 < 2.6 remission, > 5.1 high • FRAX ≥ 20% / ≥ 3%.

Note: score cut-offs and DOAC doses follow FDA labeling and current ACC/AHA, ESC, CHEST, ISTH, IDSA, AASLD, KDIGO, GOLD and GINA guidance. If your course slides or reference use slightly different cut-offs, **use your course numbers for the exam**.